Degree and Zero-Radius Clock Obstructions for Dynamical-Determinant Models of the Fibonacci Trace Map

Anonymous Authors

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Abstract

The Fibonacci Schrödinger trace map combines reversible polynomial dynamics, symbolic coding, and a canonical self-adjoint Hamiltonian, making it a natural stress test for dynamical-determinant constructions. We separate the trace-map clock k from the physical approximant length $q_k = F_{k+2}$. At coupling $\lambda = 1$, two rational escape witnesses and 48 exact modular gcd calculations refute the literal identifications of a marked spectral section with a trace-map return at times $m = k$ or q_k. In Casdagli's source-faithful ten-state large-coupling band language, with a verified unweighted six-state quotient, we obtain $u^\top(I - zA)^{-1}v = (1+z)/(1-z-z^2)$, whereas the closed-orbit zeta is $1/\det(I - zA)$. After rational continuation, the two have respectively a simple zero and a double pole at $z = -1$.

We then prove two all-level clock obstructions. First, even for arbitrary level-dependent finite dimensions N_k, ordinary k-fold products with a uniform polynomial energy-degree bound have trace, boundary, and order- k determinant coefficients of degree $O(k)$; the chronological discriminant $d_k(E)$ is monic of degree F_{k+2}. Second, at the exact finite-approximant section energies $E = 0, -1$, $d_k(E)$ grows super-exponentially in k, so $\sum_k d_k(E)z^k$ has radius zero. Thus no germ analytic at $z = 0$, including a normalized analytic Fredholm determinant or resolvent matrix element, can realize these values literally as renormalization-clock coefficients or logarithmic traces. These results do not exclude physical- time characteristic determinants, level-dependent constructions, nonanalytic operator series, or an indirect divisor correspondence with a separately defined energy map.

1 Introduction

The Fibonacci Hamiltonian is a rare setting in which a one-dimensional self-adjoint spectral problem is linked to a low-dimensional polynomial dynamical system [6]. Its trace map has a hyperbolic nonwandering set, Markov codings, periodic orbits, and thermodynamic formalism, while the associated Schrödinger operator is canonical rather than fitted [3, 5]. Trace maps also belong to a broader character-surface framework with close analogies to Hénon dynamics [2]. These features make the model a serious stress test for proposals that seek to identify a dynamical zeta or Fredholm determinant with a spectral determinant.

There is, however, a basic semantic risk. The trace map advances one *renormalization* step, whereas the corresponding periodic Schrödinger word grows by a Fibonacci substitution. A spectral condition at level k refers to an ordered product over F_{k+2} physical sites. A periodic-orbit coefficient of a trace-map zeta refers instead to a closed return after k iterates. Replacing either object by a simple averaged transition matrix erases precisely the ordered matrix product that creates the spectral polynomial.

This paper gives four exact versions of that distinction.

1. We write the marked spectral-section and periodic-return equations in the same coordinates. At coupling one, a band edge and a discriminant-zero root already have escaping trace-map orbits. A modular audit then finds gcd one in all 48 registered section/return systems, covering levels $1 \le k \le 8$, sections $d_k = 0, \pm 2$, and return clocks $m = k, q_k$. A separate implementation repeats the audit at a second prime and over $\mathbb{Q}[E]$ through level five.

2. For Casdagli's source-faithful ten-state large-coupling band language, explicit initial and terminal vectors give the marked path count $u^\top A^k v = F_{k+2}$. We compute both the boundary resolvent and the closed-orbit determinant. They share one Fibonacci pole factor but are not the same rational function or divisor. A six-state unweighted quotient is verified separately, including the required boundary lift.

3. We prove a dimension-independent all-level obstruction for passive- parameter polynomial transfer models. The state dimension may vary arbitrarily with k, but a uniform local energy-degree bound forces the k-step closed traces, uniformly degree-bounded boundary observables, and order-k determinant coefficients to have energy degree $O(k)$. The exact Fibonacci discriminant instead has degree F_{k+2}.

4. At the two exact finite-approximant section energies we prove $\lim_{k\to\infty} |d_k(E)|^{1/k} = \infty$. Consequently the renormalization-clock generating series has radius zero. This rules out literal coefficient or logarithmic-trace realization by any fixed scalar analytic germ, not only by finite matrices.

The last two statements are class theorems, not a universal Fredholm no-go. Composition operators can increase polynomial degree nonlinearly; an operator may use the physical clock q_k; a family may change with k; and a genuinely infinite-dimensional scalar matrix element may fail to be analytic at $z = 0$. More generally, a proposed divisor correspondence need not identify $d_k(E)$ with the coefficient sequence at all. We list these escape routes explicitly so that the theorems cannot be misread as excluding an undefined construction.

Prior-work boundary. Casdagli established a ten-region Markov coding and a six-symbol quotient in his stated large-coupling regime [3]; modern all-coupling spectral/dynamical relations are developed by Damanik, Gorodetski, and Yessen [5]. Reversibility, periodic orbits, and multipliers of trace maps have also been studied systematically [7]. The finite-type zeta identity is classical Bowen–Lanford theory [1]. We do not claim any of these ingredients as new. The contribution is the exact incidence audit, the source-faithful boundary/closed comparison, the dimension-independent degree–clock theorem, and the zero- radius theorem at two exact spectral-section energies.

2 Chronological Fibonacci cocycle and its two clocks

Let

$$w_{-1} = b, \qquad w_0 = a, \qquad w_{k+1} = w_k w_{k-1}. \tag{1}$$

The word lengths satisfy

$$q_{-1} = q_0 = 1, \qquad q_{k+1} = q_k + q_{k-1}, \qquad q_k = F_{k+2}, \tag{2}$$

where $F_0 = 0, F_1 = 1$.

Fix $V(a) = \lambda$, $V(b) = 0$, and define

$$A_v(E) = \begin{pmatrix} E - v & -1 \\ 1 & 0 \end{pmatrix} \in SL_2(\mathbb{C}[E]). \tag{3}$$

For a chronological word $w = c_1 \cdots c_q$, later physical sites multiply on the left:

$$M(w; E) = A_{V(c_q)}(E) \cdots A_{V(c_1)}(E). \tag{4}$$

Thus $M(uv) = M(v)M(u)$ and $M_{k+1} = M_{k-1}M_k$. We never replace this product by an averaged letter-incidence matrix.

Lemma 1 (Trace recursion). *Let $d_k(E) = \operatorname{tr} M(w_k; E)$, with the auxiliary seed $d_{-2} = 2$. Then*

$$d_{k+1} = d_k d_{k-1} - d_{k-2}, \qquad (d_{-2}, d_{-1}, d_0) = (2, E, E - \lambda). \tag{5}$$

Proof. The matrix recurrence follows from (1) and (4). Applying the standard SL_2 trace identity to the two consecutive monodromy matrices yields (5). It may also be verified directly from the seed matrices; the reproducibility package performs both checks symbolically. □

Set $x_k = d_k/2$. The recursion becomes $x_{k+1} = 2x_k x_{k-1} - x_{k-2}$, generated by

$$T(x, y, z) = (2xy - z, x, y), \qquad \ell_\lambda(E) = \left(\frac{E - \lambda}{2}, \frac{E}{2}, 1 \right), \tag{6}$$

with

$$T^k \ell_\lambda(E) = (x_k, x_{k-1}, x_{k-2}). \tag{7}$$

The Fricke–Vogt invariant $I = x^2 + y^2 + z^2 - 2xyz - 1$ obeys $I(\ell_\lambda(E)) = \lambda^2/4$.

Proposition 2 (Physical degree). *For every $k \geq -1$, $d_k(E)$ is monic and*

$$\deg_E d_k = q_k = F_{k+2}. \tag{8}$$

Proof. The assertion holds for $d_{-1} = E$, $d_0 = E - \lambda$, and directly for $d_1 = E(E - \lambda) - 2$. Starting with $k = 1$, suppose it holds at the preceding levels. In (5), the product $d_k d_{k-1}$ is monic of degree $q_k + q_{k-1} = q_{k+1}$, strictly larger than the degree q_{k-2} of the subtracted term. Its leading term cannot cancel. Induction proves the claim. □

Equation (8) exposes the two clocks used throughout: k is trace-map renormalization time, while q_k is the number of chronologically ordered lattice sites in the periodic approximant.

3 Marked spectral incidence and closed return

The length-q_k Floquet discriminant is $d_k(E)$. Let θ denote the total Floquet phase across the entire q_k-site cell, so that a Bloch solution satisfies $M_k(E)\psi = e^{i\theta}\psi$. Since $\det M_k = 1$, its spectral equation is $d_k(E) = 2\cos\theta$. Band edges have $\theta = 0, \pi$, while $\theta = \pi/2$ gives the discriminant-zero condition $d_k(E) = 0$. By (7), these are one-coordinate section incidences:

$$T^k \ell_\lambda(E) \in \{x = -1, 0, 1\}. \tag{9}$$

A return after m trace-map steps instead requires all three equations

$$d_m = d_0, \qquad d_{m-1} = d_{-1}, \qquad d_{m-2} = 2. \tag{10}$$

Neither $m = k$ nor $m = q_k$ turns (9) into (10).

Lemma 3 (Escape region)**.** *If three consecutive half traces satisfy*

$$1 < |x_{n-2}| < |x_{n-1}| < |x_n|, \tag{11}$$

then

$$|x_{j+1}| > |x_j||x_{j-1}|$$

for every $j \geq n$. In particular, there is $c > 0$ such that $\log |x_{n+r}| \geq cF_{r+2}$ for all $r \geq 0$; the orbit grows super-exponentially in trace-map time.

Proof. The reverse triangle inequality and the trace recursion give

$$|x_{n+1}| \geq 2|x_n||x_{n-1}| - |x_{n-2}| > |x_n||x_{n-1}|.$$

The last inequality follows from $|x_n||x_{n-1}| > |x_{n-2}|$. It implies the strict ordering needed to propagate (11). Set $a_r = \log |x_{n+r}|$. Then $a_{r+1} > a_r + a_{r-1}$. Taking $c = \min(a_{-1}, a_0) > 0$, induction gives $a_r \geq cF_{r+2}$. □

Proposition 4 (Two exact nonreturns)**.** *At $\lambda = 1$, the first nontrivial discriminant is*

$$d_1(E) = E(E-1) - 2. \tag{12}$$

The band edge $E = 0$, for which $d_1 = -2$, and the discriminant-zero root $E = -1$, for which $d_1 = 0$, have unbounded trace-map orbits. In particular, neither point satisfies (10) for any $m \geq 1$.

Proof. Starting at index -2, the two half-trace sequences are respectively

$$1, 0, -\tfrac{1}{2}, -1, 1, -\tfrac{3}{2}, -2, 5, -\tfrac{37}{2}, -183, \ldots$$

and

$$1, -\tfrac{1}{2}, -1, 0, \tfrac{1}{2}, 1, 1, \tfrac{3}{2}, 2, 5, \ldots.$$

Each contains the strict escape triple $(3/2, 2, 5)$ in absolute value. The escape lemma applies. □

These are exact spectral-section energies of the first finite approximant. Their escaping orbits mean, in particular, that they are not being asserted to lie in the spectrum of the infinite Fibonacci Hamiltonian.

3.1 Exact factor audit

For $c \in \{0, 2, -2\}$, define $f_{k,c} = d_k - c$. At coupling one the registered calculation forms

$$\gcd\big(f_{k,c}, d_m - d_0, d_{m-1} - d_{-1}, d_{m-2} - 2\big) \tag{13}$$

in $\mathbb{F}_{1000003}[E]$, for $1 \leq k \leq 8$ and $m \in \{k, q_k\}$. All $8 \cdot 3 \cdot 2 = 48$ gcds have degree zero.

Because $f_{k,c}$ is monic, any common factor over $\mathbb{Q}[E]$ can be chosen monic in $\mathbb{Z}[E]$. Division of the other integral polynomials by this monic factor stays integral, and reduction modulo a prime preserves its positive degree. Hence gcd one in (13) excludes every common root over $\overline{\mathbb{Q}}$ for the frozen four equations. A separate implementation, using a different polynomial representation, verifies all 48 rows at $p = 1000033$ and 30 systems directly in $\mathbb{Q}[E]$ for $k \leq 5$.

This certificate concerns the return equations (10). It cannot exclude a zero of a different energy-dependent Fredholm determinant arising from cancellations among global trace coefficients.

3.2 An explicit boundary/closed comparison

Casdagli proves a Markov coding in his $V_C \geq 8$ convention [3, Secs. 2–3]. Centering the potential gives $E_C = E - \lambda/2$, $V_C = \lambda/2$, with an exchange of the first two seed coordinates; hence this source regime corresponds to $\lambda \geq 16$ for positive coupling. Its spectral-band language uses the source-faithful ten-state graph. With rows as current states and columns as next states, define A_{10} by

$$\begin{array}{llllll} 1 \to \{3,7,10\}, & 2 \to \{4,8,9\}, & 3 \to 6, & 4 \to 5, & & \\ 5 \to 1, & 6 \to 2, & 7 \to 1, & 8 \to 1, & 9 \to 2, & 10 \to 2. \end{array} \tag{14}$$

The endpoint vectors obtained from Casdagli's equation (3.1) are

$$u_{10} = e_1 + e_6, \qquad v_{10} = e_1 + e_2 + e_3 + e_4.$$

Casdagli uses the convention $F_0 = F_1 = 1$. Translated to our standard convention, the endpoint-constrained band words of n symbols (thus $n-1$ edges) selected by u_{10}, v_{10} have count F_{n+1}. Setting $n = k+1$ gives the project convention $q_k = F_{k+2}$, and exact algebra gives

$$u_{10}^\top A_{10}^k v_{10} = F_{k+2}, \qquad u_{10}^\top (I - zA_{10})^{-1} v_{10} = \frac{1+z}{1-z-z^2}. \tag{15}$$

The closed paths instead give the Bowen–Lanford zeta

$$Z_{\mathrm{AM}}(z) = \exp\left(\sum_{n\geq 1} \frac{z^n}{n} \operatorname{tr} A_{10}^n\right) = \frac{1}{\det(I - zA_{10})}, \tag{16}$$

$$\det(I - zA_{10}) = (1+z)^2(1-z+z^2)(1-z-z^2). \tag{17}$$

Thus the marked spectral-band path count in this large-coupling coding and the closed-return zeta share the Fibonacci factor $1-z-z^2$, but they carry different boundary information. After rational continuation to $z=-1$, the boundary function has a simple zero whereas Z_{AM} has a double pole.

Casdagli also gives a six-symbol quotient. The certificate constructs its class-incidence matrix Q and verifies

$$A_{10}Q = QA_6, \qquad u_{10}^\top Q = u_6^\top, \qquad Qv_6 = v_{10}.$$

For the marked language the initial quotient symbol 6 must be decorated with its lift to the old geometric state R_6, rather than R_9 or R_{10}. Arbitrary energy-dependent weights need not descend: the full operator-quotient argument is valid when $L_{10}(E)Q = QL_6(E)$. We therefore keep the ten-state presentation as the primary spectral object.

Equation (15) counts marked admissible paths; it is not an identity for the energy polynomial $d_k(E)$. This distinction motivates, rather than proves, an energy-dependent boundary operator.

4 Two renormalization-clock obstructions

We now freeze the natural short-clock class suggested by (17)–(15). Energy is a passive parameter under ordinary matrix multiplication, and each local transition has uniformly bounded polynomial energy complexity. We allow the finite state dimension to vary arbitrarily with the level.

Definition 5 (Uniform polynomial passive-parameter family)**.** At level k, let

$$B_k(E) = \sum_{j=0}^{D} E^j B_{k,j} \in \mathrm{Mat}_{N_k}(\mathbb{C}[E]), \tag{18}$$

where $N_k < \infty$ is arbitrary and D is independent of k. The power $B_k(E)^k$ uses ordinary matrix multiplication: it does not substitute a new function of E. For a boundary observable, let $u_k(E), v_k(E) \in \mathbb{C}[E]^{N_k}$ have entry-degrees bounded uniformly by D_u, D_v.

This includes, as a special case, a fixed finite directed graph with polynomial edge weights of degree at most D: summing parallel edges gives (18). It also includes finite-dimensional matrix weights on edges after enlarging the state space. Such finite linear representations are the standard algebraic form of recognizable weighted path series [9].

Theorem 6 (Dimension-independent linear energy-degree bound)**.** *For every $k \geq 0$,*

$$\deg_E \operatorname{tr} B_k(E)^k \leq kD, \qquad \deg_E\bigl(u_k(E)^\top B_k(E)^k v_k(E)\bigr) \leq D_u + kD + D_v. \tag{19}$$

Proof. Every entry of $B_k(E)^k$ is a sum of products of k entries of $B_k(E)$. Each product has degree at most kD, and addition cannot increase the maximum degree. The trace obeys the same bound. Multiplication by the uniformly degree-bounded vectors adds at most $D_u + D_v$. The number N_k changes only the number of summands. □

Corollary 7 (Closed and boundary no-go)**.** *No uniform polynomial passive-parameter family satisfies either*

$$\operatorname{tr} B_k(E)^k = d_k(E) \quad \textit{for all sufficiently large } k, \tag{20}$$

or

$$u_k(E)^\top B_k(E)^k v_k(E) = d_k(E) \quad \textit{for all sufficiently large } k. \tag{21}$$

The same conclusion holds with $d_k(E) - 2\cos\theta$ for fixed θ.

Proof. The right-hand side is monic of degree F_{k+2} by (8); subtracting a constant does not change that degree. The left-hand sides have degree at most affine in k by (19). Since $F_{k+2}/k \to \infty$, the bounds are eventually incompatible. □

The same argument reaches formal determinant coefficients, not merely traces.

Lemma 8 (Formal determinant coefficient bound)**.** *Suppose polynomials $a_j(E)$ satisfy $\deg_E a_j \leq jD$, and set*

$$\mathcal{D}_\pm(z, E) = \exp\left(\pm \sum_{j\geq 1} \frac{a_j(E)}{j} z^j\right) = \sum_{n\geq 0} c_n^\pm(E) z^n. \tag{22}$$

Then $\deg_E c_n^\pm \leq nD$.

Proof. The coefficient of z^n is a finite sum indexed by partitions $n = \sum_{j=1}^n j m_j$. Every corresponding product contains $a_j^{m_j}$ and has energy degree at most $D \sum j m_j = Dn$. Summation cannot increase the bound. □

For the traces $a_j = \operatorname{tr} B_k(E)^j$ of a fixed level-k matrix, the minus sign in (22) gives $\det(I - zB_k(E))$, while the plus sign gives its reciprocal as a formal series. The determinant has z-degree at most N_k; its principal-minor coefficient of order n has energy degree at most nD, and vanishes for $n > N_k$.

Corollary 9 (Coefficientwise determinant no-go)**.** *No uniform polynomial passive-parameter family can satisfy*

$$[z^k]\det(I - zB_k(E))^{\pm 1} = d_k(E)$$

for all sufficiently large k.

4.1 A zero-radius obstruction beyond polynomial weights

The degree theorem uses polynomial dependence on energy. At the two exact finite-approximant section witnesses, a value-growth argument removes that hypothesis and even removes finite dimensionality from the scalar conclusion.

Theorem 10 (Zero-radius discriminant series)**.** *Fix $\lambda = 1$ and let $E_* \in \{0, -1\}$. Then*

$$\lim_{k\to\infty} |d_k(E_*)|^{1/k} = \infty. \tag{23}$$

Consequently both series

$$\sum_{k\geq 0} d_k(E_*)z^k, \qquad \sum_{k\geq 1} \frac{d_k(E_*)}{k} z^k \tag{24}$$

have radius of convergence zero.

Proof. For $E_* = 0$, the strict escape triple $(|x_3|, |x_4|, |x_5|) = (3/2, 2, 5)$ occurs; for $E_* = -1$, the same triple occurs at indices $5, 6, 7$. The escape lemma gives, for a suitable starting index n,

$$\log |x_{n+r}(E_*)| \geq cF_{r+2}$$

with $c > 0$. Since $d_k = 2x_k$ and $F_r/r \to \infty$, taking the k-th root proves (23). Division by k does not change the root growth. The Cauchy–Hadamard formula then gives radius zero for both series in (24). □

Corollary 11 (Analytic-germ no-go)**.** *At either witness energy there is no scalar germ $G(z) = \sum_{k\geq 0} g_k z^k$, analytic near $z = 0$, for which $g_k = d_k(E_*)$ for all sufficiently large k. Nor is there a normalized analytic germ Δ, with $\Delta(0) = 1$, satisfying*

$$d_k(E_*) = -k[z^k]\log\Delta(z) \tag{25}$$

for all sufficiently large k; the same holds with the opposite sign.

Proof. Cauchy estimates on any sufficiently small circle give an exponential bound on the Taylor coefficients of an analytic germ, contradicting (23). Because $\Delta(0) = 1$, it is nonzero in some disk about the origin and has an analytic logarithm there. Its coefficients again obey a Cauchy bound, and the additional factor k in (25) does not alter exponential root growth. □

Thus a fixed bounded-operator resolvent matrix element $\langle u, (I - zL)^{-1}v\rangle$, and any normalized Fredholm determinant that is analytic at the origin, cannot realize the frozen discriminant values in either literal coefficientwise sense. Finite weighted matrices with arbitrary entries finite at E_* are special cases. The conclusion is about renormalization-clock coefficients and logarithmic traces; it is not a no-go for every possible energy-to-divisor map.

4.2 Quantitative target-injection bound and a positive control

If the uniform local degree bound is relaxed to D_k, representing $d_k(E)$ by the k-step closed trace forces

$$D_k \geq \frac{F_{k+2}}{k}. \tag{26}$$

Thus a short-clock model can evade the degree theorem by inserting exponentially growing energy complexity into its local weights, but not with a uniform local rule. At $k = 30$, the exact integer bound is $D_{30} \geq \lceil 2178309/30 \rceil = 72611$.

The obstruction survives fixed higher-block recoding. Such a recoding may change the constants D, D_u, D_v, but cannot turn the affine bound (19) into Fibonacci growth. Likewise, multiplying by a fixed polynomial prefactor, making a fixed-degree change of the energy variable, or applying an E-independent scalar normalization preserves an $O(k)$ bound. Increasing N_k alone never raises these energy degrees.

There is an exact physical-clock determinant, which is a useful positive control. If $H_{k,\theta}$ is the q_k-site periodic approximant with total Floquet phase θ, then for $q_k \geq 3$, in the monic convention,

$$\det(EI - H_{k,\theta}) = d_k(E) - 2\cos\theta. \tag{27}$$

Indeed, both sides are monic of degree q_k, and their zeros are the same Floquet energies with multiplicity. This construction uses a determinant whose order grows with the physical length; it is not the order-k coefficient of a trace-map dynamical determinant.

4.3 Escape routes beyond the frozen claims

The hypotheses deliberately leave several routes open.

1. A physical-time model may use q_k site steps or the full characteristic determinant (27). The ordinary Schrödinger monodromy does exactly this.

2. A level-dependent family may inject the local degree required by (26), change its operator space with k, or otherwise fail to define one fixed analytic germ in the variable z.

3. A composition operator can increase polynomial degree exponentially. For example,

$$(Cf)(E) = f(E^2), \qquad C^k(E) = E^{2^k}.$$

   This is linear on a function space but is not a passive-parameter family of the form (18). If its proposed scalar generating series has the coefficients in (24), that series is necessarily nonanalytic at the origin.

4. The trace-map Koopman operator also evades the degree theorem:

$$(\delta_{\ell_\lambda(E)} \circ U_T^k)(\pi_x) = \pi_x(T^k \ell_\lambda(E)) = x_k(E).$$

   Here the moving evaluation functional carries unbounded energy information; the associated scalar series at the escape witnesses has radius zero.

5. A formal zero-radius series, an unbounded operator whose relevant matrix element is not analytic at $z = 0$, or another non-Fredholm construction is outside the analytic-germ corollary.

6. An indirect divisor correspondence may use a separately defined map between z, energy, and spectral phase instead of equating d_k with trace-map coefficients. Such a candidate must freeze that map before it can be tested.

These are genuine changes of class or observable. The degree theorem applies exactly to ordinary passive-parameter products with uniformly bounded local polynomial degree; the zero-radius theorem applies exactly to literal renormalization-clock coefficient and logarithmic-trace realizations by an analytic germ.

5 Operator scope and Route-A disposition

The experiments and theorems above concern distinct candidates and parameter regimes. The incidence and zero-radius witnesses use $\lambda = 1$. The source-faithful Casdagli coding uses $\lambda \geq 16$, and we freeze $\lambda = 16$ for the unweighted Route-A record. The degree theorem is coupling-independent. Evidence is not transferred between these regimes without an explicit theorem.

5.1 Three objects that must remain separate

HCS-C13-AM: unweighted Artin–Mazur zeta. For the frozen hyperbolic coding,

$$Z_{\mathrm{AM}}(z) = \exp\left(\sum_{n\geq 1} \frac{z^n}{n} \operatorname{tr} A_{10}^n\right) = \det(I - zA_{10})^{-1} \tag{28}$$

is an intrinsic closed-return function, normalized by $Z_{\mathrm{AM}}(0) = 1$. For the direct candidate only, we freeze the literal formal-series identity

$$\log Z_{\mathrm{AM}}(z) = \sum_{k\geq 1} \frac{d_k(E)}{k} z^k,$$

equivalently $\operatorname{tr} A_{10}^k = d_k(E)$ in $\mathbb{C}[E]$ for every k. This is refuted: $\operatorname{tr} A_{10}^k$ is an integer independent of energy, whereas $d_k(E)$ is a monic polynomial of degree F_{k+2}. The boundary series (15) is a different marked object. No alternative map from z to energy or to a target Riemann divisor is implicit in (28).

HCS-C13P and HCS-C13G: scoped repair obstructions. HCS-C13P permits arbitrary finite dimensions N_k but freezes a uniform local polynomial energy-degree bound; the dimension-independent theorem refutes literal trace, boundary, and order-k determinant-coefficient repairs in that class. HCS-C13G removes polynomiality at the two exact finite-approximant section energies and refutes literal coefficient or logarithmic-trace matching by every fixed scalar germ analytic at $z = 0$. These are proved obstruction records, not additional evidence layers of HCS-C13-AM.

HCS-C13R: general energy-dependent Fredholm proposal. A schematic expression such as

$$\left\langle u, (I - z\mathcal{L}_{E,\theta})^{-1} v\right\rangle \tag{29}$$

does not define a mathematical candidate. A function space, operator domain, energy and phase dependence, weights, clock, determinant convention, normalization, and boundary functionals are required. Zeros of a defined Fredholm determinant could arise through global cancellations and need not make the marked point $\ell_\lambda(E)$ periodic. This is compatible with the standard Ruelle determinant framework [8]. HCS-C13R is therefore rejected at the Route-A input gate as `NOT_TESTABLE`; it is not assigned A1–A4 scores and is not a universal Fredholm no-go.

5.2 The natural Hamiltonian is not a discrete spectral lift

The infinite Fibonacci Hamiltonian is self-adjoint, but that fact alone does not provide a Hilbert–Pólya lift. Sütő proved zero-measure Cantor spectrum in the Fibonacci model [10], and Damanik–Lenz proved absence of eigenvalues for every element of the Fibonacci hull [4]. Hence the natural infinite operator does not supply a discrete eigenvalue sequence. This excludes that operator itself, not every new operator that one might derive from the trace dynamics.

5.3 Route-A screening of the frozen unweighted claim

Table 1 scores only HCS-C13-AM at $\lambda = 16$, with trace-map period as its clock and (28) as its determinant convention. No prime or Riemann-zero data, fitting, or cross-regime incidence data enter this disposition.

Layer	Verdict	Reason
A1	`A1_WEAK`	Intrinsic primitive trace-map orbits exist, but no canonical prime correspondence, logarithmic prime clock, or arithmetic repetition weights are supplied.
A2	`A2_FAIL`	The frozen literal coefficient claim is refuted: $\operatorname{tr} A_{10}^k$ is energy-independent, while $d_k(E)$ has degree F_{k+2}. No alternative frozen z-to-energy divisor map is defined.
A3	`A3_FAIL`	No completed-ξ normalization, functional equation, gamma factor, Euler weights, target divisor, or counting law is present.
A4	`A4_FORMAL_HINT`	A canonical self-adjoint Hamiltonian exists, but it uses the physical lattice clock, has no point spectrum, and is not a quantization of (28).

Table 1: Route-A screening of HCS-C13-AM. The overall status is `ROUTE_A_REJECTED` for this direct unweighted construction.

Route B is not invoked. HCS-C13P and HCS-C13G remain reusable scoped obstructions; HCS-C13R can be reconsidered only after its full operator tuple and target map are defined. The research decision is to stop local tuning of the unweighted Fibonacci candidate and switch to a system whose arithmetic clock and ordered cocycle are intrinsic from the outset.

6 Conclusion

The Fibonacci trace map contains genuine symbolic, periodic-orbit, and self-adjoint spectral structures, but their determinants are not interchangeable. A finite approximant discriminant is produced by a marked chronological product over F_{k+2} sites. A trace-map zeta coefficient is a closed return after k renormalization steps. The exact section/return audit and the source-faithful ten-state boundary calculation expose this semantic difference.

Two all-level results make the clock mismatch quantitative. Under a uniform local polynomial energy-degree bound, traces, marked boundary powers, and order-k determinant coefficients have degree $O(k)$, even if the finite state dimension N_k grows arbitrarily. The exact discriminant degree is F_{k+2}. At the two rational finite-approximant section witnesses, the discriminant values also grow so rapidly that their renormalization-clock series has radius zero, excluding literal realization by any fixed analytic germ at the origin.

The correct research consequence is a switch, not further local tuning. A future Fibonacci construction must visibly use physical time, growing-order characteristic determinants, level-dependent data, a nonanalytic operator mechanism, or a fully defined indirect divisor map. Otherwise the next exploration should move to a dynamical system whose arithmetic clock and ordered cocycle are intrinsic from the start.

A Reproducibility certificate

All calculations use exact integer, rational, symbolic, or finite-field arithmetic. No Riemann prime or zero data are read. From the project root, run

```
python code/trace_map_audit.py --out-dir results --max-k 8 --prime 1000003
python code/coding_boundary_audit.py --out-dir results --max-k 15
python code/degree_clock_audit.py --out-dir results --max-k 30 \
  --max-symbolic-k 10
python code/value_growth_audit.py --out-dir results --through-k 18
python code/independent_check.py
python code/test_trace_map_audit.py
python code/test_coding_boundary_audit.py
python code/test_degree_clock_audit.py
python code/test_value_growth_audit.py
```

The principal artifacts are listed in Table 2.

Artifact	Exact content
`results/certificate.json`	recurrence, invariant, chronological products, and two rational escape witnesses
`results/modular_gcd_audit.csv`	48 producer-prime section/return gcds
`results/independent_check.json`	48 persisted-prime, 48 second-prime, and 30 rational gcd checks, plus schema and direct matrix checks
`results/symbolic_boundary_certificate.json`	source-faithful ten-state determinant and boundary series, plus the decorated six-state quotient
`results/degree_clock_certificate.json`	dimension-independent theorem scope, symbolic degrees through $k = 10$, and explicit escape routes
`results/degree_growth.csv`	integer lower bound $\lceil F_{k+2}/k \rceil$ through $k = 30$
`results/value_growth_certificate.json`	exact rational escape triples, propagated product inequalities, and the zero-radius theorem scope

Table 2: Machine-readable exact artifacts.

The independent checker does not import the producer. It uses a separate coefficient-array polynomial implementation, direct 2×2 chronological matrix multiplication, a second prime 1000033, and direct rational gcds through $k = 5$. The ordinary unit tests are labeled regression tests and are not presented as independent evidence.

References

[1] Rufus Bowen and III Lanford, Oscar E. Zeta functions of restrictions of the shift transformation. In *Global Analysis*, volume 14 of *Proceedings of Symposia in Pure Mathematics*, pages 43–49. American Mathematical Society, 1970. doi: 10.1090/pspum/014/9985.

[2] Serge Cantat. Bers and hénon, painlevé and schrödinger. *Duke Mathematical Journal*, 149(3): 411–460, 2009. doi: 10.1215/00127094-2009-042.

[3] Martin Casdagli. Symbolic dynamics for the renormalization map of a quasiperiodic schrödinger equation. *Communications in Mathematical Physics*, 107(2):295–318, 1986. doi: 10.1007/BF01209396.

[4] David Damanik and Daniel Lenz. Uniform spectral properties of one-dimensional quasicrystals, i. absence of eigenvalues. *Communications in Mathematical Physics*, 207(3):687–696, 1999. doi: 10.1007/s002200050742.

[5] David Damanik, Anton Gorodetski, and William Yessen. The fibonacci hamiltonian. *Inventiones Mathematicae*, 206(3):629–692, 2016. doi: 10.1007/s00222-016-0660-x.

[6] Mahito Kohmoto, Leo P. Kadanoff, and Chao Tang. Localization problem in one dimension: Mapping and escape. *Physical Review Letters*, 50(23):1870–1872, 1983. doi: 10.1103/PhysRevLett.50.1870.

[7] John A. G. Roberts and Michael Baake. Trace maps as 3d reversible dynamical systems with an invariant. *Journal of Statistical Physics*, 74:829–888, 1994. doi: 10.1007/BF02188581.

[8] David Ruelle. Zeta-functions for expanding maps and anosov flows. *Inventiones Mathematicae*, 34:231–242, 1976. doi: 10.1007/BF01403069.

[9] Marcel-Paul Schützenberger. On the definition of a family of automata. *Information and Control*, 4(2–3):245–270, 1961. doi: 10.1016/S0019-9958(61)80020-X.

[10] András Sütő. Singular continuous spectrum on a cantor set of zero lebesgue measure for the fibonacci hamiltonian. *Journal of Statistical Physics*, 56(3–4):525–531, 1989. doi: 10.1007/BF01044450.